



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	COMPUTATIONAL CHEMISTRY				
Type of Course	ELECTIVE IN MAJOR				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. A solid understanding of fundamental chemistry principles, including atomic structure, chemical bonding, molecular geometry, and chemical reactions. 2. Knowledge of basic physics principles, particularly classical mechanics and electromagnetism, as well as an understanding of quantum mechanics at an introductory level.				
Course Summary	This course provide students with theoretical knowledge and practical skills in using computational methods to solve chemical problems. The course covers a range of topics, including introduction to computational methods, molecular modeling, and molecular dynamics simulations, and emphasizes hands-on experience with computational tools and software.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category #	Evaluation Tools used
CO1	Understand the history, scope, definitions, and fundamentals of computational chemistry including computational methods and Molecular modelling Techniques.	U	F	Instructor-created exams / Quiz /Assignment
CO2	Apply a variety of computational methods used in chemistry, such as molecular mechanics (MM), and Monte Carlo	Ap	C	Class test /Assignment /Quiz

	simulations, to solve chemical problems and analyze molecular systems.			
CO3	Effectively utilize computational software package Gaussian commonly employed in chemical research, to perform calculations, analyze data, and visualize molecular structures and properties.	U	P	Assignment/ Class test
CO4	Interpret computational results obtained from simulations and calculations, including molecular structures, energetics, spectroscopic properties, and reaction mechanisms, and relate them to experimental observations.	An	C	Assignments /Seminar presentation
CO5	Explore applications of computational chemistry in various fields, such as drug discovery, materials science, catalysis, and environmental chemistry, and understand how computational methods contribute to advancing scientific research and solving real-world problems.	Ap	M	Assignments /Seminar presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Computational Chemistry		9	15
	1	Theory, computation & modeling – Definition of terms.	2	
	2	Need of approximate methods in quantum mechanics. Computable Quantities – structure, potential energy surfaces and chemical properties.	3	
	3	Cost & Efficiency – relative CPU time, software & hardware.	2	
	4	Classification of computational methods.	2	
II	Computer Simulation Methods- I		9	15

	6	Introduction – molecular dynamics and Monte Carlo methods.	2	
	7	Calculation of simple thermodynamic properties - energy, heat capacity, pressure and temperature, phase space, practical aspects of computer simulation.	3	
	8	Periodic boundary conditions, Monitoring the equilibration.	2	
	9	Analyzing the results of a simulation, error estimation.	2	
III	Computer Simulation Methods- II		12	20
	10	Molecular dynamics (MD) method – molecular dynamics using simple models	2	
	11	MD with continuous potentials, finite difference methods, choosing the time step, setting up and running a MD simulation.	3	
	12	Monte Carlo (MC) method - calculating properties by integration, .	3	
	13	Metropolis method, random number generators.	2	
	14	MC simulation of rigid molecules.	2	
IV	ab initio Methods and basic concepts of Density Functional Theory in Computational Chemistry		18	30
	15	Review of Hartree – Fock method for atoms, SCF treatment of polyatomic molecules.	2	
	16	Closed shell systems - restricted HF calculations; Open shell systems – ROHF and UHF calculations; The Roothan – Hall equations, Koopmans theorem	2	
	17	HF limit & electron correlation, Introduction to electron correlation (post -HF) methods	3	
	18	Basics of DFT- Applications of computational chemistry in various fields, such as drug discovery, materials science, catalysis, and environmental chemistry, and understand how computational methods contribute to advancing scientific research and solving real-world problems.	3	
	19	Hydrogen-like, Slater-type & Gaussian type basis functions, classification of basis sets – minimal, double zeta, triple zeta, split-valence, polarization & diffuse basis sets, even tempered & well-tempered basis sets, contracted basis sets,	3	

	20	Pople-style basis sets and their nomenclature, correlation consistent basis sets, basis set truncation error, effect of choice of method/ basis set (model chemistries) on cpu time.	3	
V	Representation of Molecular Geometry and Gaussian Calculations (Open Ended)		12	20
	Open Ended	Specification of molecular geometry using a) Cartesian coordinates and b) Internal coordinates. The Z-matrix, Z-matrices of some simple molecules like H ₂ , H ₂ O, formaldehyde ammonia and methanol. Simple calculations using Gaussian programme	12	

References:

1. C. J. Cramer, Essentials of computational Chemistry: Theories and models, John Wiley & Sons 2002.
2. Frank Jensen, Introduction to Computational Chemistry, John Wiley & Sons LTD1999.
3. J. Foresman & Aelieen Frisch, Exploring Chemistry with Electronic Structure Methods, Gaussian Inc., 2000.
4. David Young, Computational Chemistry- A Practical Guide for Applying Techniques to Real- World Problems”, Wiley -Interscience, 2001.
5. Errol G. Lewars, Computational Chemistry: Introduction to the theory and applications of molecular quantum mechanics, 2 nd edn, Springer2011.
6. I.N. Levine, Quantum Chemistry, 6th Edition, Pearson Education Inc., 2009.
7. P.W. Atkins & R.S. Friedman, Molecular quantum mechanics, 4th Edition, Oxford University Press, 2005.
8. W. Koch, M.C. Holthausen, “A Chemist’s Guide to Density Functional Theory”, Wiley VCH Verlag2000.

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	1					3		1	1	1		1
CO 2	2	2					3		2	2	2		2
CO 3		3					3		2	2	2		3
CO 4		3					3		2	2	2		3
CO 5		3					3		2	2	2		3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓